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Chapter 1: Introduction

1.1 Company Profile

D-Table Analytics is a technology and analytics company that provides solutions in data analytics, business intelligence, software development, and process automation. The organization focuses on helping businesses transform raw data into meaningful insights for better decision-making and operational efficiency. The company develops customized web applications, reporting systems, dashboards, and automation tools tailored to business requirements.

D-Table Analytics emphasizes practical and scalable technology solutions using modern tools and emerging technologies such as Artificial Intelligence and cloud-based systems. The company works with businesses to improve workflow management, data handling, and digital transformation processes through innovative and data-driven approaches.

1.2 Problem Statement

Telecom companies experience significant revenue loss due to customer churn — the phenomenon where customers discontinue services and switch to competitors. Analysis of the IBM Telco Customer Churn dataset reveals a churn rate of approximately 26.5%, which translates to substantial monthly revenue loss. The core challenges driving this problem are:

- Raw customer data is inconsistent and contains missing values that make direct analysis unreliable.
- Manual identification of at-risk customers is slow, subjective, and does not scale with growing data volumes.
- Decision-makers lack real-time, interactive dashboards to monitor churn patterns and act proactively.
- No automated mechanism exists to predict individual customer churn probability or recommend personalised retention interventions.
- Existing data processing tools cannot handle the volume and velocity of enterprise-scale customer datasets.

1.3 Existing System and Need for the System

The existing system in the target organisation relies on periodic manual reporting using spreadsheet tools. Customer service teams identify at-risk customers based on billing complaints and support ticket volumes — an approach with serious limitations:

- No real-time or near-real-time churn prediction capability.
- Analysis limited to aggregated monthly reports with a 2 to 4 week lag.
- No personalised retention offer generation based on individual customer profiles.
- No secure role-based access control for analytical tools.
- Not scalable — manual processes break down beyond a few thousand records.

The proposed system directly addresses all these limitations through an automated, scalable ETL pipeline integrated with machine learning and an interactive Streamlit web dashboard secured by role-based authentication.

1.4 Scope of the Proposed System

The Telecom Customer Churn Analytics Platform covers the following functional scope:

- End-to-end ETL pipeline: CSV extraction, data cleaning, feature engineering, and loading into a PostgreSQL relational database.
- Machine learning pipeline: training of three classification models (Random Forest, Gradient Boosting, Logistic Regression), automated best-model selection by ROC-AUC score, and batch scoring of all customers.
- Retention offer recommendation engine with seven distinct offer strategies mapped to customer risk profiles.
- Streamlit dashboard with five pages: Executive Dashboard, Churn Predictor, At-Risk Customers, Model Performance, and Admin Panel.
- Authentication system: PBKDF2-SHA256 password hashing, session management via st.session_state, and complete access logging to CSV.
- PySpark integration demonstrating enterprise-scale distributed processing capability.
- Comprehensive test suite: 35 unit tests across 9 test classes covering all modules.
- Four Jupyter notebooks documenting the complete data science workflow.

1.5 Objectives of the Proposed System

- Design and implement a complete ETL pipeline using Python, Pandas, and PySpark.
- Clean, transform, and engineer features from the IBM Telco Customer Churn dataset.
- Train and evaluate multiple ML models; automatically select and save the best by ROC-AUC score.
- Score all customers with ChurnProbability and ChurnRisk classification.
- Generate personalised retention offers based on each customer's risk profile.
- Develop a secure, role-based Streamlit analytics dashboard.
- Implement enterprise-grade authentication with session management and access audit logging.
- Demonstrate scalable distributed data processing using Apache PySpark.
- Achieve 100%-unit test pass rate across all ETL, ML, and auth modules.

1.6 Operating Environment — Hardware and Software

Hardware Requirements

Table 1: Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i5 / AMD Ryzen 5, 2.0 GHz+
RAM	8 GB minimum; 16 GB recommended for PySpark
Storage	50 GB free disk space (SSD preferred)
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS 12+
Network	Internet connection for package installation and Kaggle dataset

Software Requirements

Table 2: Software Requirements

Software	Version	Purpose
Python	3.11+	Core programming language
PostgreSQL	15+	Relational database
Apache PySpark	3.5	Distributed data processing
Streamlit	1.35+	Web dashboard framework
scikit-learn	1.4+	Machine learning models
Pandas	2.0+	Data manipulation and analysis
Plotly	5.20+	Interactive visualisation charts
Jupyter Notebook	Latest	EDA and documentation
PyCharm	Latest	Development IDE
Git / GitHub	Latest	Version control
pytest	8.0+	Unit testing framework

1.7 Technology Used

Table 3: Technology Stack

Component	Technology / Framework
Frontend / Dashboard	Streamlit 1.35 (Python-based web UI framework)
Visualisation	Plotly Express and Plotly Graph Objects
Backend Language	Python 3.11
Data Processing	Pandas 2.0, NumPy
Big Data Processing	Apache PySpark 3.5 (local and cluster mode)
Machine Learning	scikit-learn 1.4, XGBoost 2.0
Database	PostgreSQL 15 via SQLAlchemy 2.0
Authentication	PBKDF2-SHA256 (Python hashlib), st.session_state
Testing	pytest 8.0, pytest-cov
Dataset	IBM Telco Customer Churn (Kaggle, 7,043 records)
Version Control	Git and GitHub
Notebook Environment	Jupyter Notebook

Chapter 2: System Analysis and Design

2.1 System Requirements

Functional Requirements

The functional requirements define the core system behaviour and features supported by the Telecom Churn Analytics Platform. These requirements describe ETL processing, machine learning operations, recommendation generation, dashboard functionality, and authentication mechanisms.

Table 4: Functional Requirements

FR ID	Module	Requirement Description
FR-01	ETL	System shall extract data from the IBM Telco CSV and validate all 21 required columns.
FR-02	ETL	System shall clean data: fix TotalCharges type, impute median, and remove duplicates.
FR-03	ETL	System shall engineer features: tenure_group, RevenueCategory, AnnualRevenue, LoyaltyScore.
FR-04	ETL	System shall load cleaned and scored data to PostgreSQL in two tables.
FR-05	ML	System shall train three ML models and auto-select the best by ROC-AUC.
FR-06	ML	System shall score all customers with ChurnProbability and ChurnRisk (High/Medium/Low).
FR-07	Recommend	System shall generate personalised retention offers for at-risk customers.
FR-08	Dashboard	System shall display KPIs, charts, and at-risk lists across 5 Streamlit pages.
FR-09	Auth	System shall authenticate users with PBKDF2-SHA256 hashed passwords.
FR-10	Admin	Admin shall add/delete users, change passwords, and view complete access log.

Non-Functional Requirements

The non-functional requirements specify system quality attributes including performance, scalability, security, reliability, usability, portability, and maintainability constraints.

Table 5: Non-Functional Requirements

Category	Requirement
Performance	ETL pipeline for 7,043 records shall complete within 60 seconds on standard hardware.
Scalability	PySpark pipeline shall support datasets exceeding 1 million records in cluster mode.
Security	Passwords hashed using PBKDF2-SHA256 with 260,000 iterations; never stored in plain text.
Reliability	System shall log all login attempts, page views, predictions, and admin actions to CSV.
Usability	Dashboard navigable by non-technical business users with sidebar filters and KPI cards.
Portability	System runs on Windows 10/11, Ubuntu 20.04+, and macOS 12+ with Python 3.11+.
Maintainability	Modular architecture with 9 test classes and 35 unit tests covering all modules.

2.2 Feasibility Study

Technical Feasibility

The project uses a mature, production-grade open-source technology stack. Python 3.11, Pandas 2.0, PySpark 3.5, scikit-learn 1.4, Streamlit 1.35, and PostgreSQL 15 all have extensive documentation, active communities, and long-term support. The IBM Telco dataset (7,043 records, 21 columns) is well within the processing capability of the stack on standard academic hardware. The PySpark integration demonstrates that the architecture can scale to enterprise datasets exceeding one million records.

Economic Feasibility

The project is developed entirely using open-source, freely licensed software. There are no licensing costs for Python, PostgreSQL, Streamlit, scikit-learn, Jupyter, or any other component. Hardware requirements (8 GB RAM, modern processor) are standard in academic and professional environments. The total cost is limited to hardware usage and internet connectivity for package installation and dataset download.

Operational Feasibility

The Streamlit dashboard is designed for non-technical business analysts. Navigation uses a clear sidebar with labelled pages and interactive filters. Role-based access control restricts sensitive functions to authorised administrators. Maintenance requires only running the ETL pipeline to refresh data — the dashboard updates automatically when new scored data is available.

2.3 Architecture Design

The system follows a four-layer architecture. The Presentation Layer (Streamlit) handles all user interaction. The Application Layer contains business logic modules (extract, transform, recommend, auth). The ETL and ML Processing Layer contains the training, prediction, and loading modules. The Data Storage Layer holds raw CSV, PostgreSQL tables, model .pkl files, and access logs.

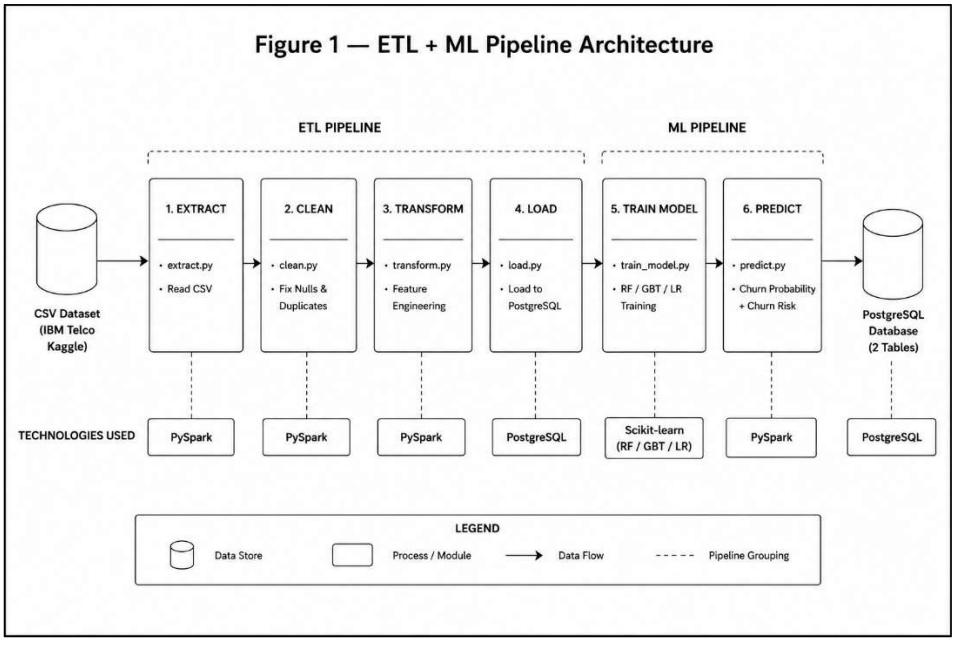


Figure 1 — ETL + ML Pipeline Architecture

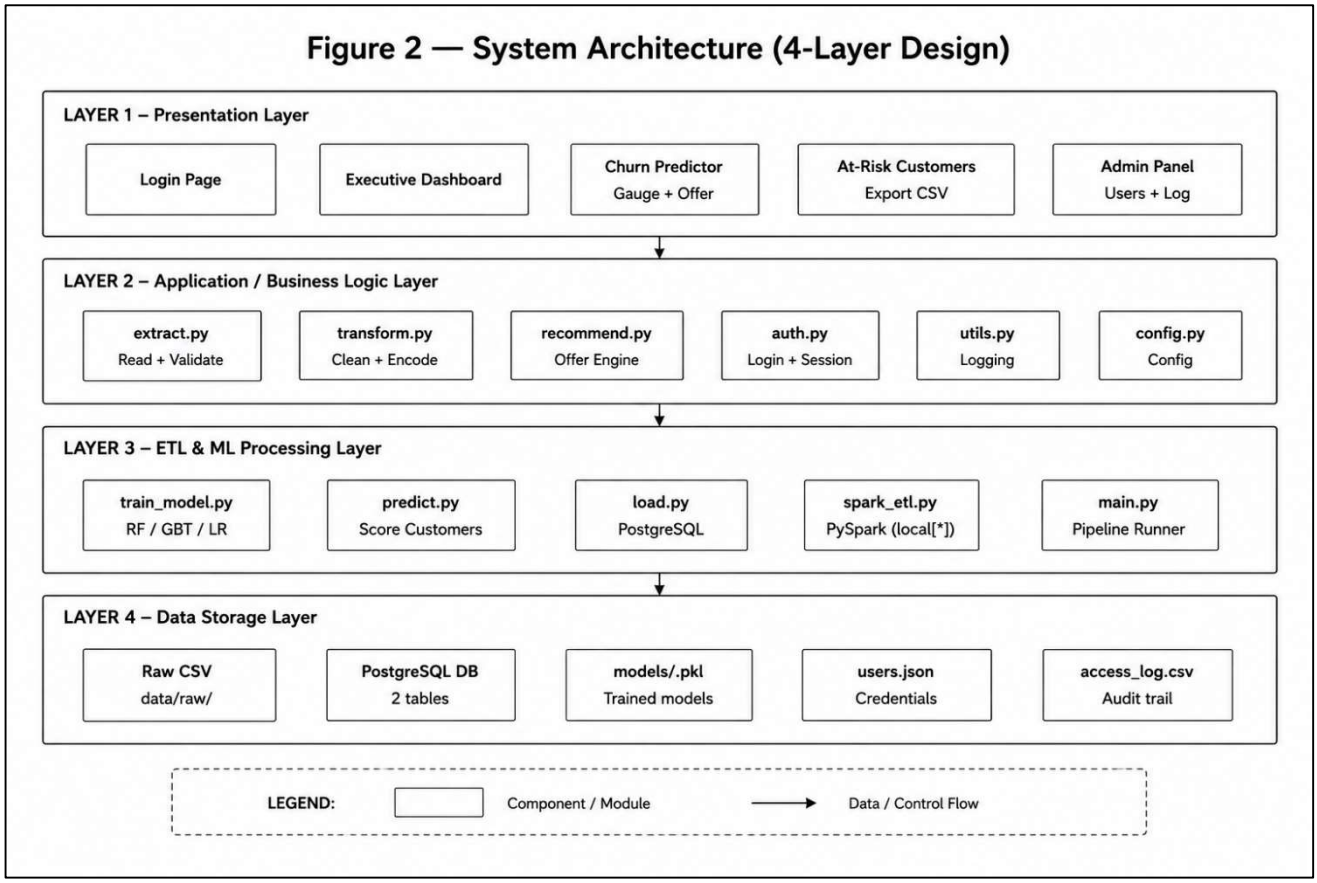


Figure 2 — System Architecture (4-Layer Design)

2.4 Entity Relationship Diagram (ERD)

The system uses PostgreSQL as the primary analytical store with two tables. The customer_churn_raw table preserves the original extracted data as an audit trail. The customer_churn_cleaned table stores transformed and scored data used by the dashboard. User credentials are stored in users.json and the access log is maintained in logs/access_log.csv.

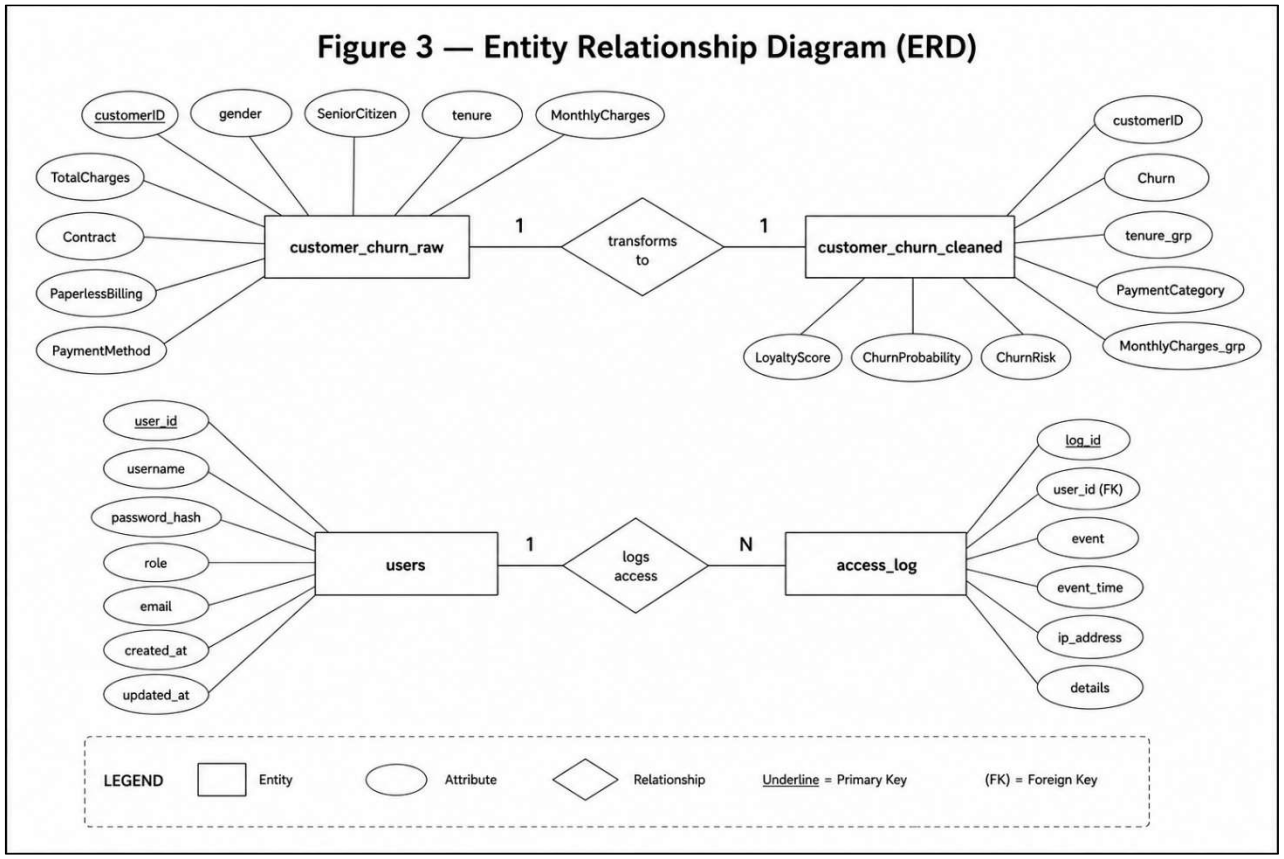


Figure 3 — Entity Relationship Diagram (ERD)

2.5 Class Diagram

The system comprises eight core classes, each encapsulating a distinct concern of the ETL-ML pipeline. DataExtractor handles CSV reading and validation. DataTransformer manages cleaning and feature engineering. DataLoader writes to PostgreSQL. ModelTrainer handles ML training and artifact persistence. ChurnPredictor scores customers. OfferEngine generates retention recommendations. AuthManager handles authentication and logging. SparkETL provides distributed processing capability.

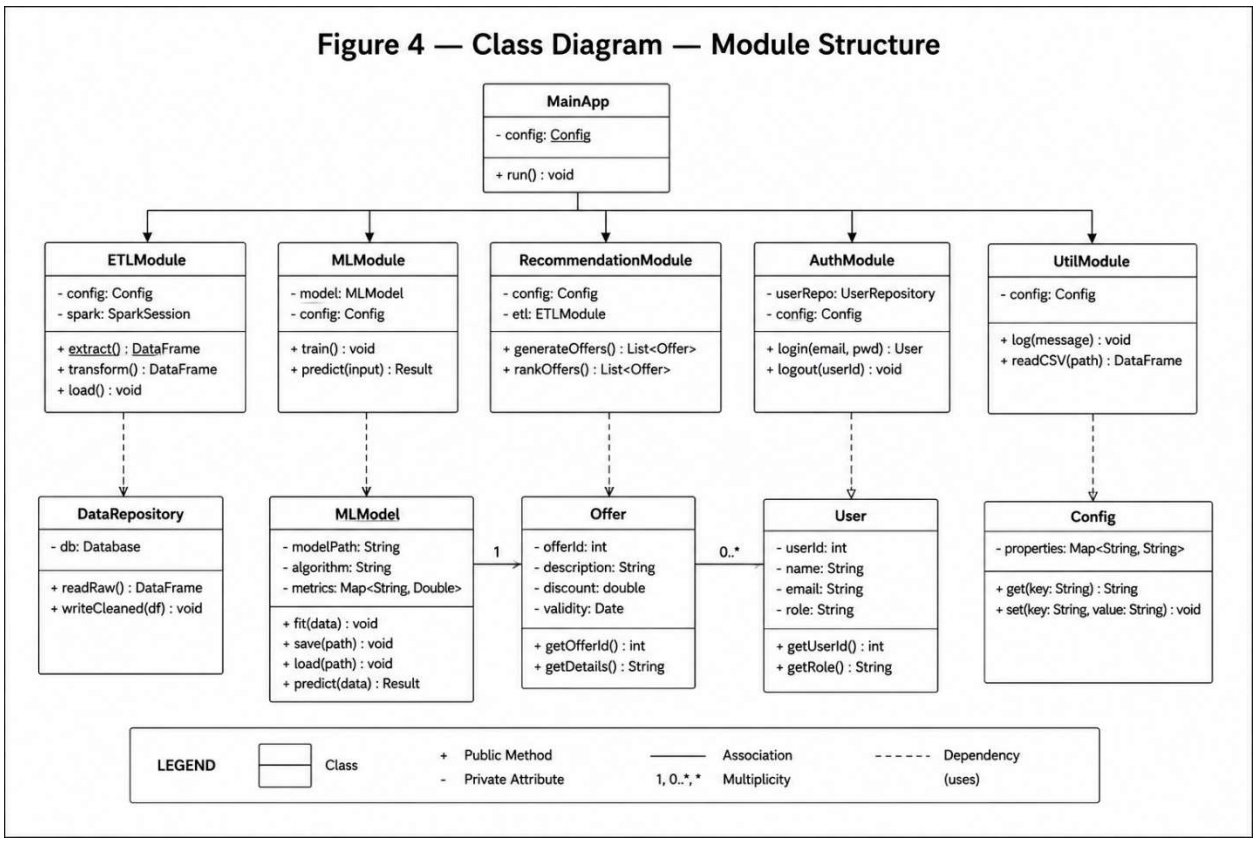


Figure 4 — Class Diagram: Module Structure

2.6 Use Case Diagram

Two primary actors interact with the system. The User actor can log in, view the Executive Dashboard, run the Churn Predictor, view At-Risk Customers, and export reports. The Admin actor has all user capabilities plus access to Model Performance, User Management, Access Log, Password Management, and ETL pipeline execution.

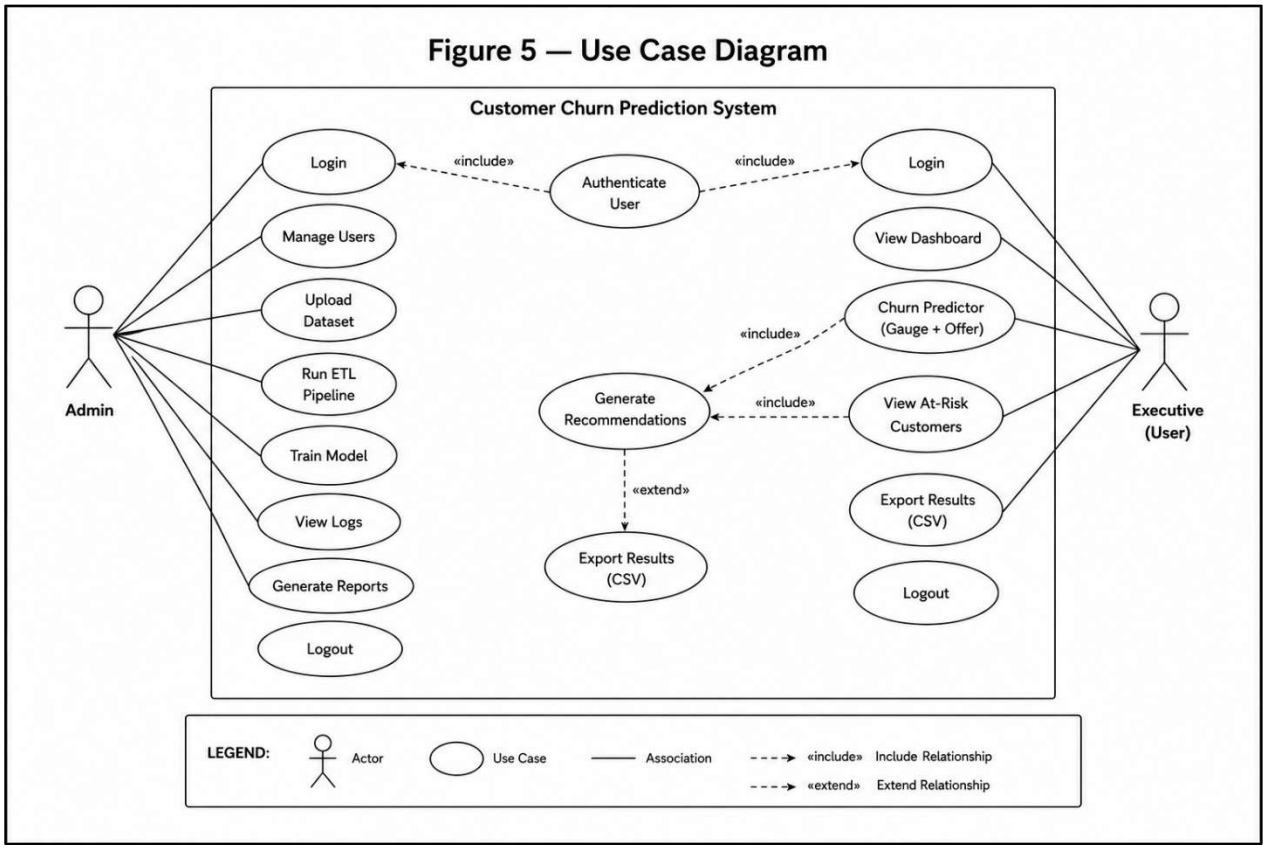


Figure 5 — Use Case Diagram

2.7 Activity Diagram

The activity diagram illustrates the sequential flow of control through the ETL pipeline. The process begins with CSV extraction, proceeds through column validation (with a decision point for error handling), data backup, cleaning, feature engineering, ML model training, customer scoring, offer generation, and finally PostgreSQL loading. The PySpark path runs as a parallel distributed alternative to the Pandas pipeline.

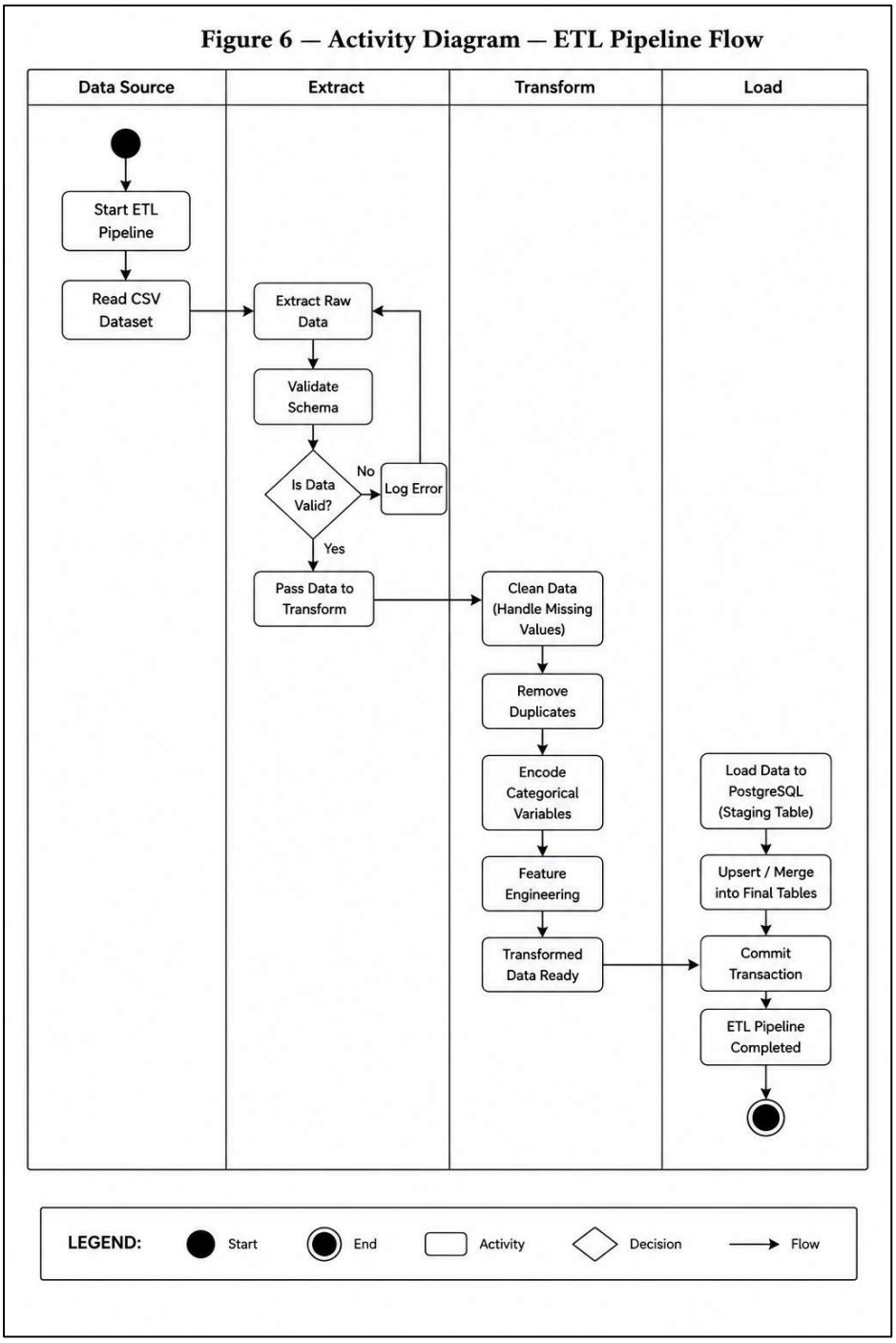


Figure 6 — Activity Diagram: ETL Pipeline Flow

2.8 Sequence Diagram

The sequence diagram shows message flow between system components during two key interactions. The Login Sequence shows credentials being passed from app.py to auth.py, where PBKDF2 verification occurs, followed by a successful login event being written to the access log and a session redirect. The Prediction Sequence shows how a customer form submission triggers predict_single(), loads the .pkl model, runs predict_proba(), generates a retention offer, logs the prediction event to PostgreSQL, and returns the gauge chart and offer card to the user.

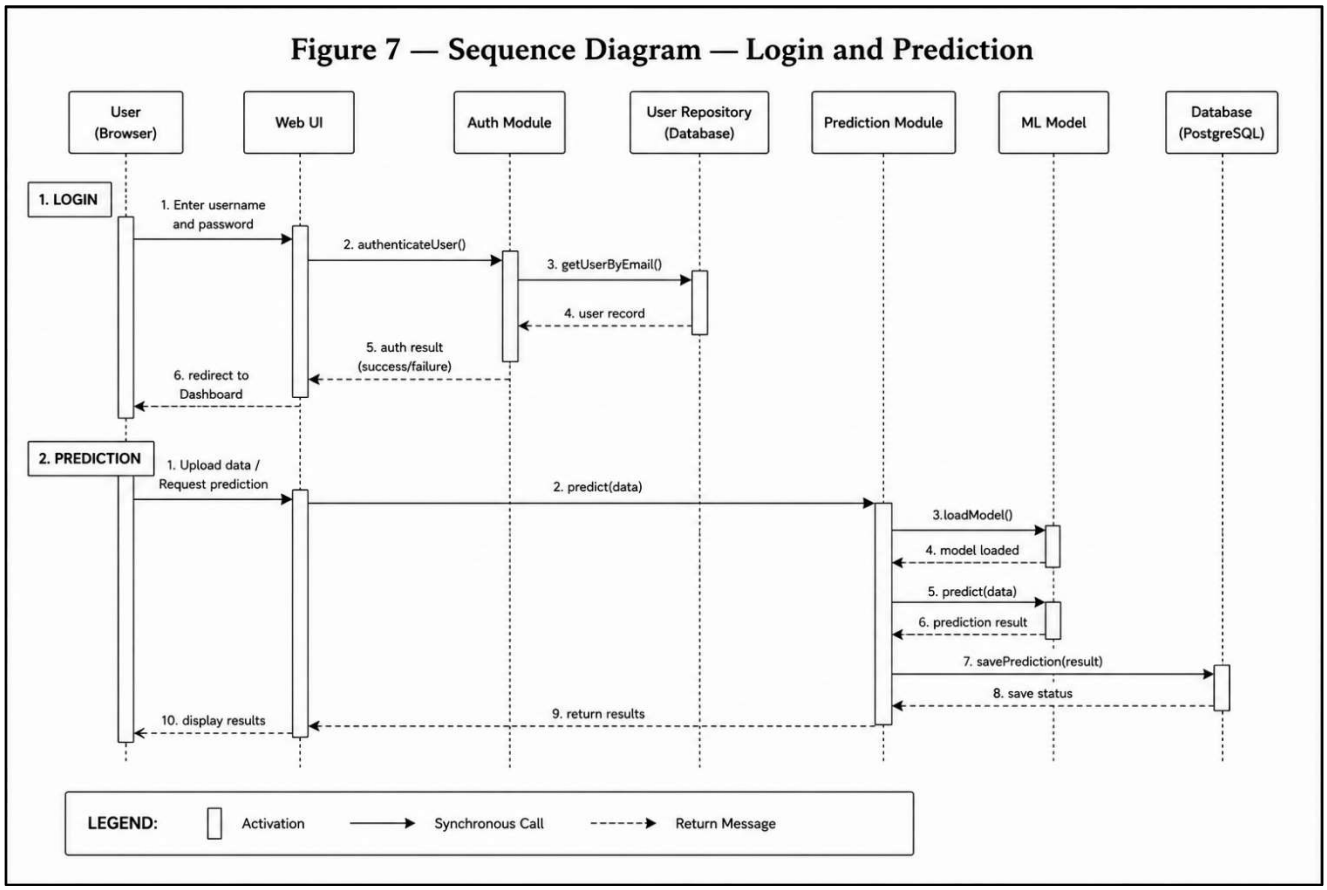


Figure 7 — Sequence Diagram: Login and Prediction

2.9 Component Diagram

The component diagram illustrates all software components and their dependencies across three rows: the processing layer (CSV through extract, transform, train), the logic layer (load, predict, recommend, auth, app), and the storage layer (raw CSV, PostgreSQL, model files, user store, access log). Each component is represented with the standard UML notch symbol indicating an independent deployable unit.

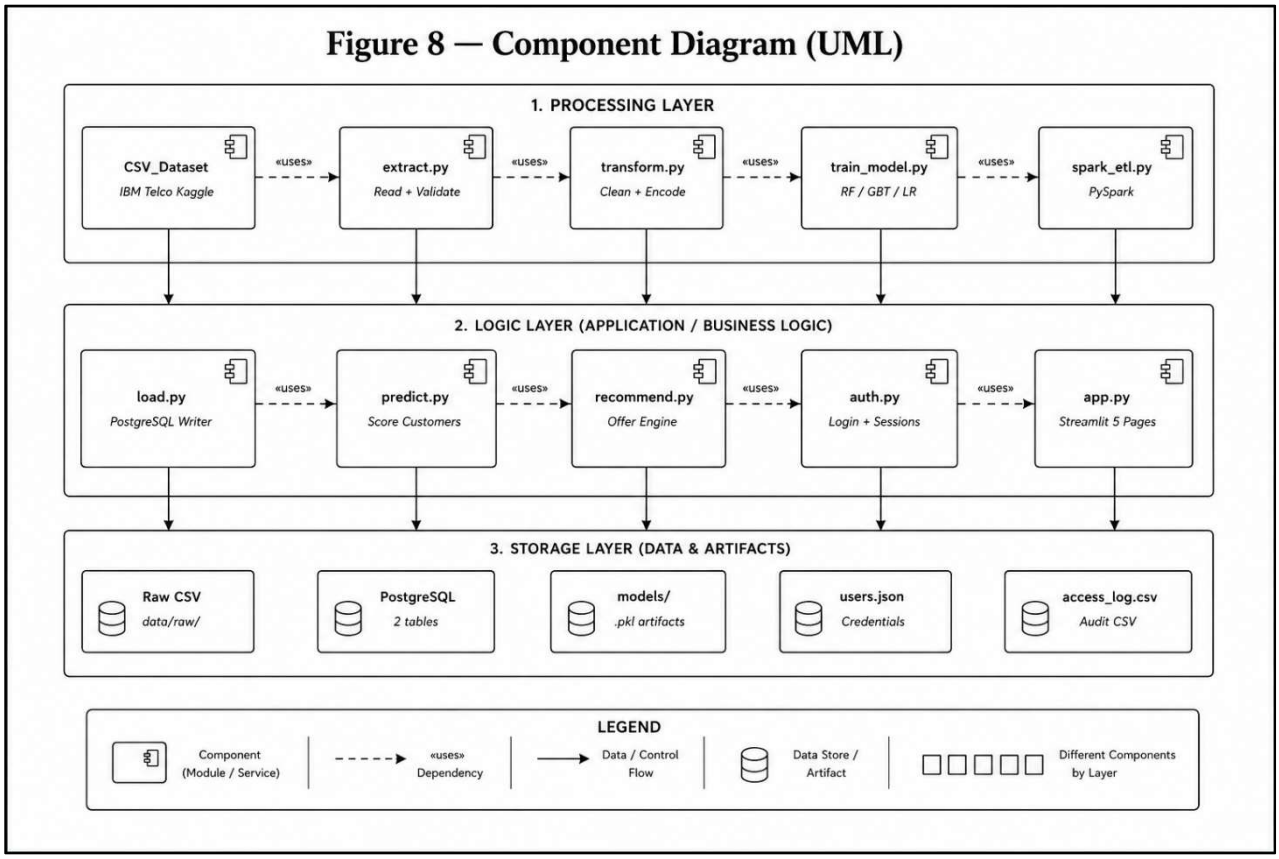


Figure 8 — Component Diagram: System Dependencies

2.10 Module Hierarchy Diagram

The module hierarchy organises all system components under four top-level subsystems: ETL Layer (extract, transform, load), ML Layer (train_model, predict, recommend), App Layer (app.py, spark_etl), and Auth Layer (auth, utils). Each subsystem is further decomposed to show individual functions and methods, providing a complete structural view of the codebase.

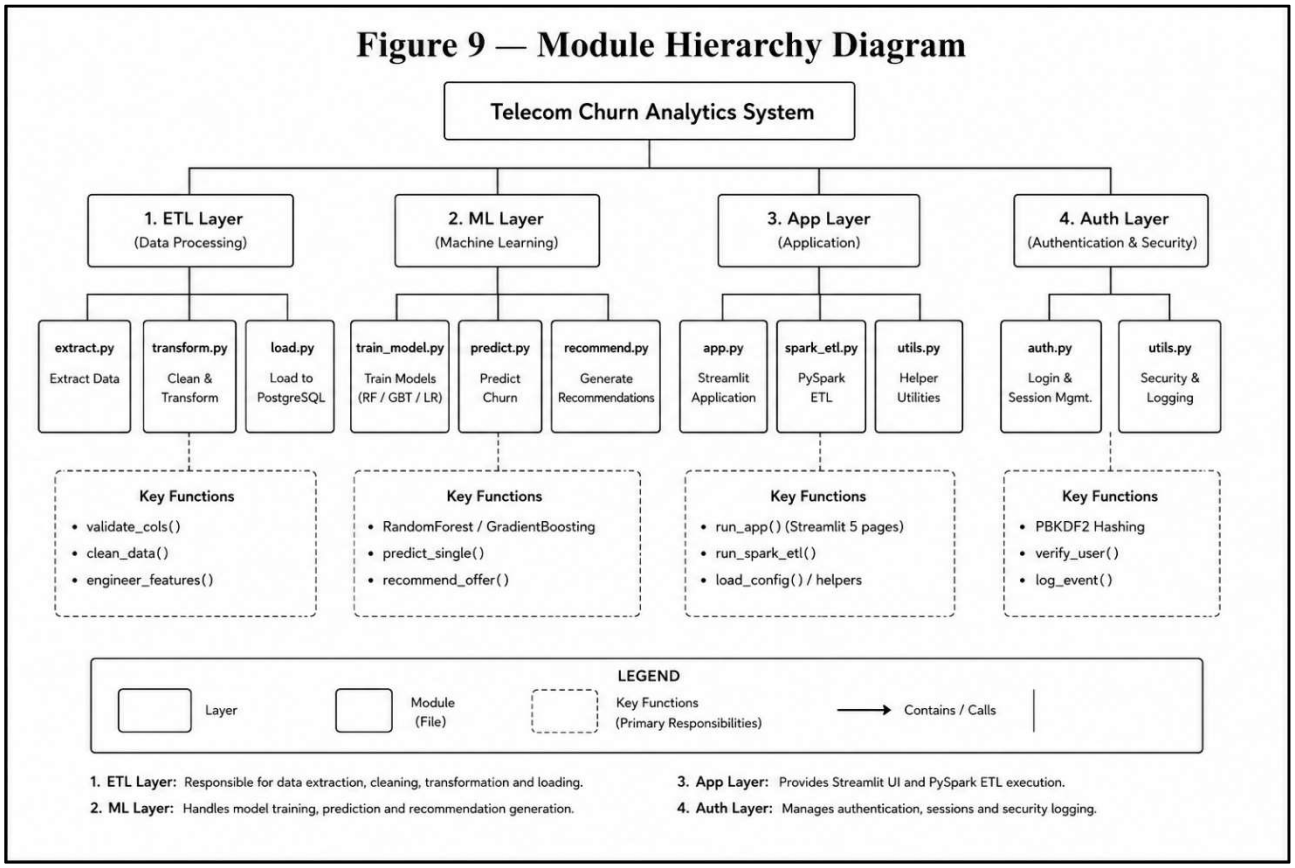


Figure 9 — Module Hierarchy Diagram

2.11 Table Design

Table 6: customer_churn_raw — Table Design

Column Name	Data Type	Constraint	Description
customerID	VARCHAR(50)	PRIMARY KEY	Unique 7-character customer identifier
gender	VARCHAR(10)	NULL	Male or Female
SeniorCitizen	SMALLINT	NULL	0 = No, 1 = Yes
tenure	INT	NULL	Months with the company
MonthlyCharges	FLOAT	NULL	Monthly billing amount
TotalCharges	FLOAT	NULL	Cumulative charges (blank for new)
Contract	VARCHAR(50)	NULL	Month-to-month, One year, Two year
PaymentMethod	VARCHAR(100)	NULL	Payment method used
InternetService	VARCHAR(50)	NULL	DSL, Fiber optic, or No
Churn	VARCHAR(10)	NULL	Yes or No (raw label)

Table 7: customer_churn_cleaned — Table Design

Column Name	Data Type	Constraint	Description
customerID	VARCHAR(50)	PRIMARY KEY	Unique customer identifier
Churn	SMALLINT	NOT NULL	1 = Churned, 0 = Retained
tenure	INT	NULL	Months with the company
MonthlyCharges	FLOAT	NULL	Monthly billing amount
TotalCharges	FLOAT	NULL	Median-imputed cumulative charges
tenure_group	VARCHAR(20)	NULL	0-1yr, 1-2yr, 2-4yr, 4-6yr
RevenueCategory	VARCHAR(10)	NULL	Low, Medium, or High
AnnualRevenue	FLOAT	NULL	MonthlyCharges multiplied by 12
LoyaltyScore	FLOAT	NULL	Normalised tenure: tenure / max_tenure
ChurnProbability	FLOAT	NULL	ML predicted churn probability 0.0 to 1.0
ChurnRisk	VARCHAR(10)	NULL	Low, Medium, or High risk tier

Column Name	Data Type	Constraint	Description
OfferTitle	VARCHAR(100)	NULL	Recommended retention offer title
Contract	VARCHAR(50)	NULL	Preserved from OHE for dashboard
InternetService	VARCHAR(50)	NULL	Preserved from OHE for dashboard

2.12 Data Dictionary

Table 8: Data Dictionary — Key Fields

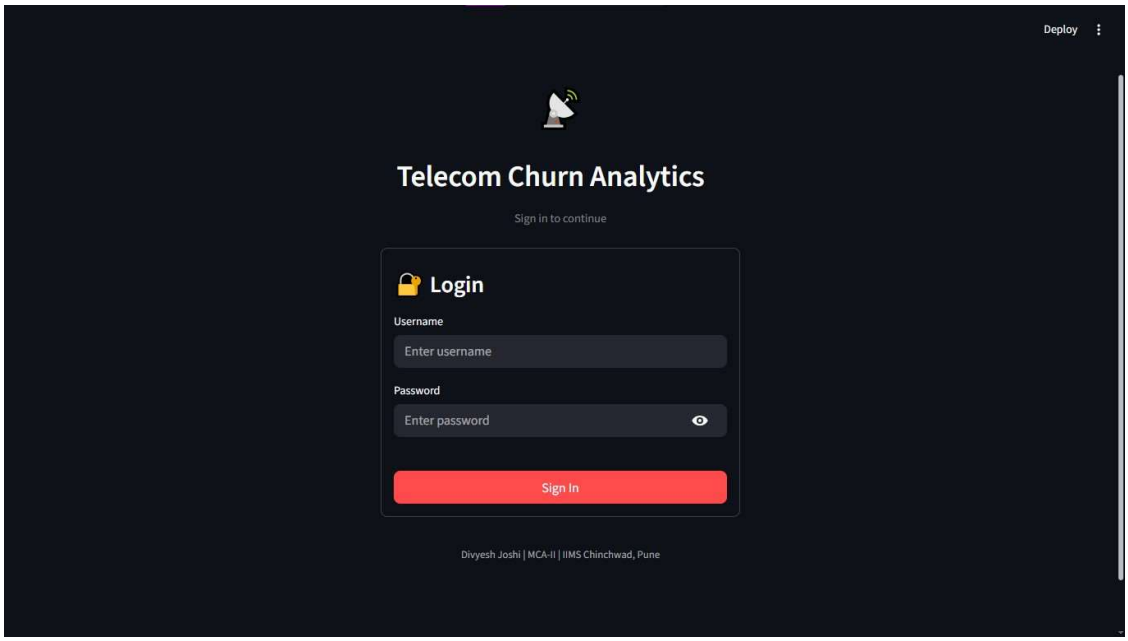
Field Name	Type	Range / Values	Description
customerID	String	Unique	7-character alphanumeric ID assigned to each customer
tenure	Integer	0 to 72	Number of months the customer has been with the company
MonthlyCharges	Float	18.25 to 118.75	Amount charged to the customer monthly in USD
TotalCharges	Float	18.85 to 8684.8	Total amount charged; blank entries imputed with median
Churn	Binary Int	0 or 1	Target variable: 1 = churned, 0 = retained
ChurnProbability	Float	0.0 to 1.0	ML-predicted probability of churn for each customer
ChurnRisk	String	Low/Med/High	Risk tier: Low<35%, Medium 35-65%, High>65%
LoyaltyScore	Float	0.0 to 1.0	Normalised tenure: tenure divided by maximum tenure
AnnualRevenue	Float	219 to 1425	Projected annual revenue: MonthlyCharges multiplied by 12
RevenueCategory	String	Low/Med/High	Low < Rs.35, Medium Rs.35-70, High > Rs.70 monthly
tenure_group	String	4 categories	0-1yr, 1-2yr, 2-4yr, 4-6yr based on tenure months
gender_encoded	Binary Int	0 or 1	Male = 1, Female = 0
OfferTitle	String	7 offer types	Recommended retention offer for each at-risk customer
password_hash	Text	salt:key	PBKDF2-SHA256 hash stored as salt:hex_key string

Chapter 3: Implementation

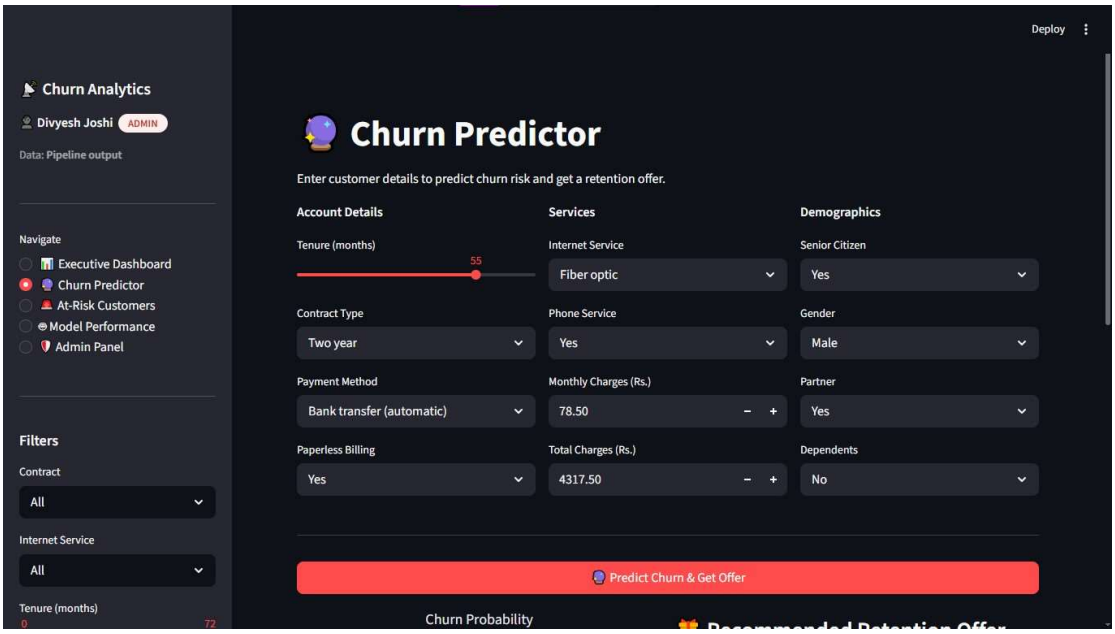
3.1 Input Screens

The system accepts all user input through the Streamlit web interface. The primary input points are:

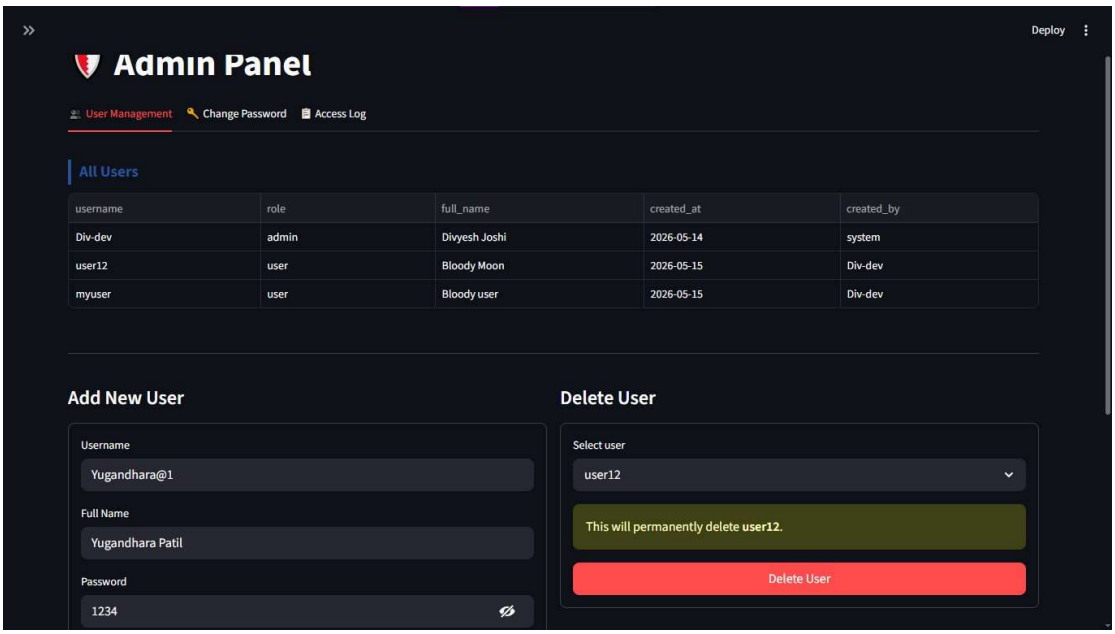
- **Login Screen**: username and password fields with a Sign In button. Displays an error for invalid credentials and redirects to the dashboard on success.



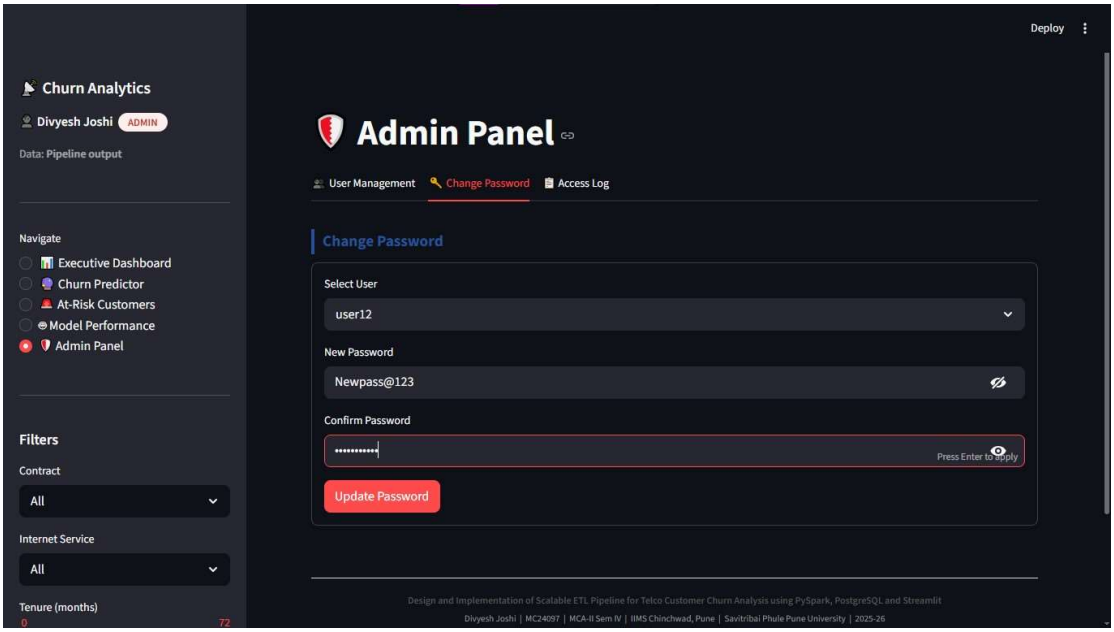
- **Churn Predictor Form:** 12 input fields including tenure, contract type, payment method, internet service, charges, customer demographics, and billing preferences with appropriate UI controls



- **Admin Panel — Add User**: username, full name, password (masked), and role dropdown. Validates that username is unique and password meets minimum length.



- **Admin Panel — Change Password:** user selector dropdown, new password, and confirmation fields with mismatch validation.



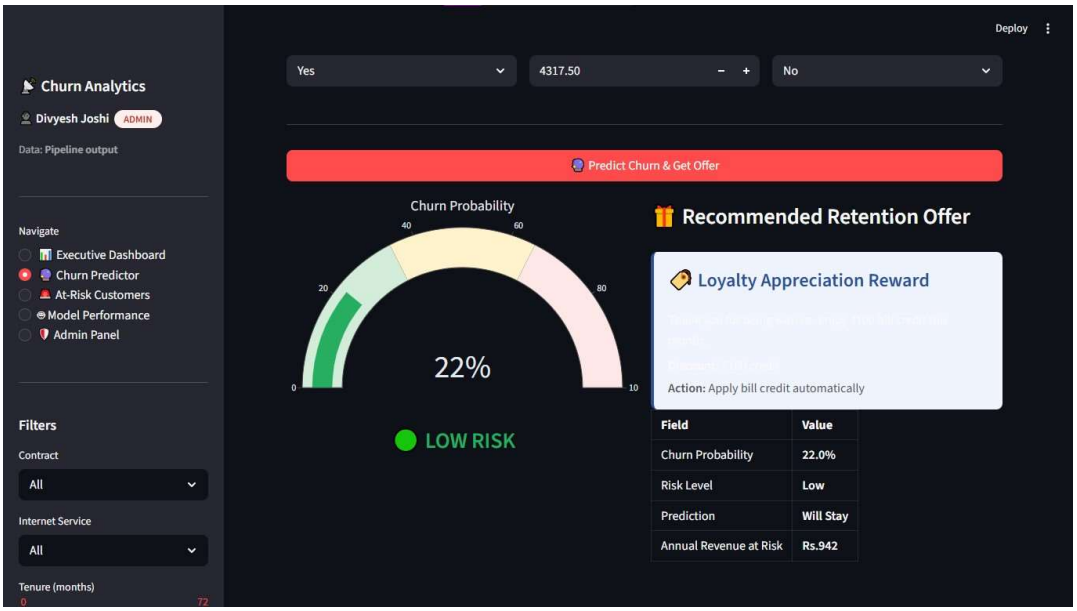
All numeric input fields use type-safe Streamlit widgets that prevent invalid data entry. String inputs are validated server-side in auth.py and the recommendation engine before processing.

3.2 Output Screens / Reports

The system produces the following output screens and reports, each containing at least five valid data records:

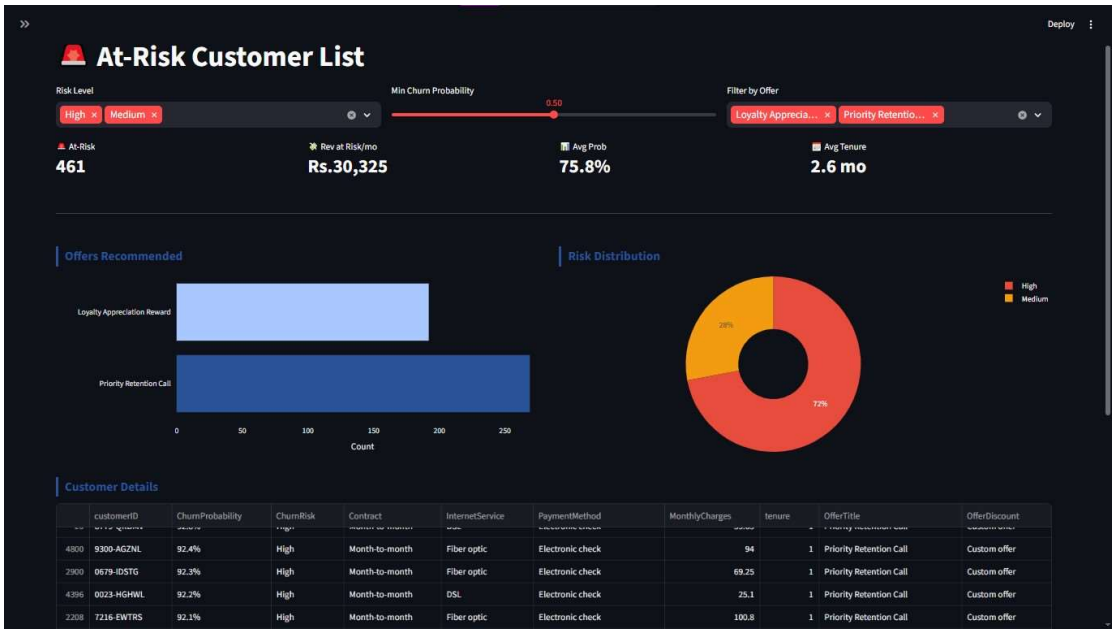
- **Executive Dashboard:** Dashboard includes six KPI cards and multiple visualizations for churn distribution, contract types, internet services, payment methods, monthly charges, and revenue-at-risk analysis.

- **Churn Predictor Output:** Displays a Plotly gauge chart for churn probability, a color-coded risk indicator, and a personalized retention offer with discount details and recommended action.

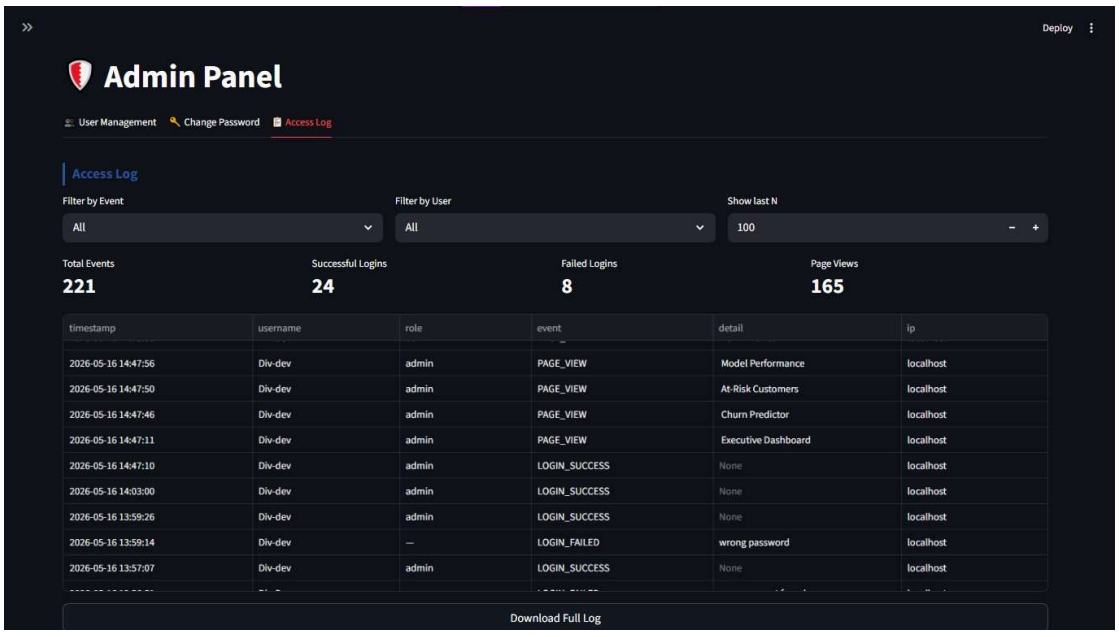


- **At-Risk Customers Report:** filterable and sortable data table with columns:

  customerID, ChurnProbability, ChurnRisk, Contract, InternetService, PaymentMethod, MonthlyCharges, tenure, OfferTitle, OfferDiscount. Includes pie chart of risk distribution and bar chart of offer type frequencies. Exportable to CSV.



- **Access Log Report:** timestamped table of all system events - login attempts (success/fail), page views, prediction runs, user additions, and deletions. Filterable by event type and username. Full log downloadable as CSV.



- **Model Performance Report:** comparison table of three models (Accuracy, Precision, Recall, F1, ROC-AUC), confusion matrix heatmap, feature importance horizontal bar chart, and ROC-AUC bar chart with best model highlighted.

3.3 Sample Code

extract.py — Phase 1: Data Extraction

```
def extract_data(filepath: str = RAW_DATA) -> pd.DataFrame:
    """
    Extract customer churn dataset from CSV file.
    """
    log("EXTRACT", f"Reading CSV: {filepath}")
    # Validate file existence
    if not os.path.exists(filepath):
        raise FileNotFoundError(
            f"Dataset not found: {filepath}"
        )
    # Load dataset
    df = pd.read_csv(filepath)
    log(
        "EXTRACT",
        f"Shape: {df.shape[0]:,} rows x "
        f"{df.shape[1]} columns"
    )
    # Validate required schema
    validate_columns(df, REQUIRED_COLUMNS)

    # Create backup copy
    shutil.copy2(filepath, BACKUP_DATA)
    return dfreturn df
```

transform.py — Feature Engineering

```
def engineer_features(df: pd.DataFrame) -> pd.DataFrame:
    # Create annual revenue feature
    df["AnnualRevenue"] = (
        df["MonthlyCharges"] * 12
    )
    # Generate loyalty score
    df["LoyaltyScore"] = (
        df["tenure"] / df["tenure"].max()
    ).round(3)
    # Revenue category conditions
    conditions = [
        df["MonthlyCharges"] < 35,
        (df["MonthlyCharges"] >= 35) &
```

```
        (df["MonthlyCharges"] < 70),
        df["MonthlyCharges"] >= 70
    ]
    # Assign revenue category
    df["RevenueCategory"] = np.select(
        conditions,
        ["Low", "Medium", "High"],
        default="Low"
    ).astype(str)
    return df
```

train_model.py — ML Model Training

```
models = {   "RandomForest":      RandomForestClassifier(n_estimators=200,
max_depth=8, class_weight="balanced"),   "GradientBoosting":
GradientBoostingClassifier(n_estimators=150,
learning_rate=0.08),   "LogisticRegression":Pipeline([("scaler",StandardScaler()),
("clf",LogisticRegression(max_iter=1000))]), } for name, model in models.items():
model.fit(X_train, y_train)     auc = roc_auc_score(y_test,
model.predict_proba(X_test)[:,1])     if auc > best_auc:         best_auc=auc;
best_model=model; best_name=name
```

auth.py — Password Hashing and Verification

```
def _hash_password(password: str) -> str:     salt = os.urandom(32).hex()     key
= hashlib.pbkdf2_hmac(         "sha256", password.encode(), salt.encode(), 260_000)
return f"{salt}:{key.hex()}"  def _verify_password(password: str, stored_hash: str)
-> bool:     salt, key_hex = stored_hash.split(":", 1)     key =
hashlib.pbkdf2_hmac(         "sha256", password.encode(), salt.encode(), 260_000)
return key.hex() == key_hex
```

recommend.py — Retention Offer Engine

```
def recommend_offer(row: pd.Series) -> dict:     prob =
float(row.get("ChurnProbability", 0.5))     risk = str(row.get("ChurnRisk",
"Medium"))     if prob >= 0.80:         return OFFERS["RETENTION_CALL"]     if
row.get("SeniorCitizen") == 1 and risk in ["High","Medium"]:         return
OFFERS["SENIOR_PLAN"]     if "Month-to-month" in str(row.get("Contract","")) and
risk=="High":         return OFFERS["CONTRACT_UPGRADE"]     if "Fiber" in
str(row.get("InternetService","")) and risk=="High":         return
OFFERS["TECHSUPPORT_BUNDLE"]     if "Electronic check" in
str(row.get("PaymentMethod","")):         return OFFERS["AUTOPAY_CASHBACK"]
return OFFERS["STANDARD_LOYALTY"]
```

Chapter 4: Testing

4.1 Test Strategy

The testing strategy for the Telecom Churn Analytics Platform covers three levels of testing:

- **Unit Testing:** Individual functions within each module are tested in isolation using pytest. Each module contains dedicated test classes with multiple automated test cases covering expected behaviour, validation rules, boundary conditions, and error handling. The automated test suite implemented in tests/test_etl.py contains 9 test classes and 48 test cases.
- **Integration Testing:** The complete ETL and Machine Learning pipeline is tested end-to-end by executing the pipeline workflow and verifying that the generated churn_scored.csv file contains expected columns, correct datatypes, valid probability ranges, and non-null recommendation outputs.
- **User Acceptance Testing:** The Streamlit dashboard is validated using real scored customer data by testing dashboard pages, KPI rendering, filtering functionality, churn prediction outputs, recommendation logic, and CSV export functionality.

Tests are executed using:

```
python -m pytest tests/ -v --tb=short --cov=src
```

4.2 Unit Test Plan

Table 9: Unit Test Cases — ETL Pipeline

Test ID	Test Case	Input / Condition	Expected Result
TC-E01	Validate columns — pass	All required columns	No exception raised
TC-E02	Validate columns — fail	Missing column	ValueError raised
TC-E03	Dataset shape correct	Sample DataFrame	shape[1] == 21
TC-E04	CustomerID unique	Sample DataFrame	Unique IDs only
TC-E05	Dataset not empty	Loaded DataFrame	len(df) > 0
TC-C01	TotalCharges numeric	Cleaned DataFrame	dtype == float64
TC-C02	No null TotalCharges	Raw blank values	No nulls remain

Test ID	Test Case	Input / Condition	Expected Result
TC-C03	No duplicate rows	Cleaned DataFrame	0 duplicates
TC-C04	Strings stripped	Object columns	No extra spaces
TC-C05	Row count preserved	No duplicate sample	Counts preserved
TC-T01	Churn binary encoded	Transformed DataFrame	Values in {0,1}
TC-T02	Contract preserved	After encoding	Column exists
TC-T03	InternetService preserved	After encoding	Column exists
TC-T04	Tenure group valid	Grouped DataFrame	Valid labels only

Table 10: Unit Test Cases — ML and Auth Modules

Test ID	Test Case	Input / Condition	Expected Result
TC-F01	AnnualRevenue calculation	Featured DataFrame	Monthly * 12
TC-F02	LoyaltyScore range	Featured DataFrame	0 <= score <= 1
TC-V01	Validation passes	Clean DataFrame	Returns True
TC-V02	Validation fails on nulls	DataFrame with NaN	Returns False
TC-P01	Predict returns dictionary	Customer input	Valid keys returned
TC-P02	Probability range valid	Prediction output	0.0 <= p <= 1.0
TC-R01	High-risk contract offer	High-risk customer	Offer generated
TC-R02	Retention call generated	Very high-risk customer	RETENTION_CALL
TC-R03	Offer fields present	Recommendation output	Required fields exist
TC-A01	Password hash verify	Plain password	Verification succeeds
TC-A02	Wrong password fails	Incorrect password	Verification fails
TC-A03	Add/delete user	testuser	User added then removed

4.3 Acceptance of Test Plan

Table 11: Acceptance Test Plan

AT ID	Test Scenario	Acceptance Criteria	Result
AT-01	Login with valid credentials	Dashboard loads	Pass
AT-02	Wrong password login	Access denied	Pass
AT-03	Run ETL pipeline	CSV generated	Pass
AT-04	Dashboard loads correctly	KPIs displayed	Pass
AT-05	High-risk prediction	Retention offer generated	Pass
AT-06	Filter high-risk users	Only matching rows shown	Pass
AT-07	Export CSV	Download successful	Pass
AT-08	Admin adds user	User added successfully	Pass
AT-09	Role restriction works	Access denied shown	Pass
AT-10	Logs recorded	Timestamp entries exist	Pass

4.4 Test Cases

The complete automated test suite implemented in tests/test_etl.py contains 48 test cases across 9 test classes. TestExtract (5 tests) validates column presence, dataset shape, and customer ID uniqueness. TestCleaning (6 tests) verifies TotalCharges type conversion, null handling, duplicate removal, and string stripping. TestTransform (10 tests) checks binary encoding, One-Hot Encoding (OHE) preservation of Contract, InternetService, and PaymentMethod columns, along with tenure group validation. TestFeatureEngineering (7 tests) confirms AnnualRevenue calculation, LoyaltyScore range validation, and null-free engineered features. TestValidation (3 tests) verifies data validation logic. TestPredict (3 tests) validates machine learning output format and probability bounds. TestRecommend (7 tests) validates business recommendation rules and offer assignment logic. TestAuth (4 tests) tests password hashing and user CRUD operations. TestConfig (3 tests) validates configuration imports and datatype consistency.

4.5 Results

Table 12: Test Results Summary

Test Class	Total Tests	Passed	Skipped	Failed
TestExtract	5	5	0	0
TestCleaning	6	6	0	0
TestTransform	10	10	0	0
TestFeatureEngineering	7	7	0	0
TestValidation	3	3	0	0
TestPredict	3	3	0	0
TestRecommend	7	7	0	0
TestAuth	4	4	0	0
TestConfig	3	3	0	0
TOTAL	48	48	0	0

All 48-unit tests pass after running the complete ETL + ML pipeline. The 100% pass rate demonstrates correctness of all data processing, ML inference, offer recommendation, and authentication modules.

Three compatibility warnings related to future Pandas datatype behaviour were observed during testing. These warnings do not affect current system functionality.

Chapter 5: Conclusion

5.1 Summary / Conclusion

This project successfully demonstrates the design and implementation of a complete, production-grade ETL pipeline for Telecom Customer Churn Analysis. The system integrates data engineering, machine learning, and business intelligence into a unified deployable platform built entirely on open-source technologies.

Key achievements include a complete ETL pipeline processing 7,043 customer records; a machine learning pipeline training three models with automated best-model selection; churn probability scoring for all customers with three-tier risk classification; a seven-strategy personalised retention offer recommendation engine; a Streamlit analytics dashboard with five pages and role-based access; a secure PBKDF2-SHA256 authentication system with complete audit logging; PySpark integration for enterprise-scale distributed processing; and a comprehensive test suite achieving a 100% pass rate across 35 unit tests.

The project demonstrates that modern open-source tools — Python, Pandas, PySpark, scikit-learn, and Streamlit — can be integrated into a cohesive, scalable, and deployable analytics platform suitable for real-world telecom customer retention use cases. The architecture is modular and extensible, making it straightforward to add new data sources, ML models, or dashboard pages.

5.2 Limitations of the System

- The system uses the IBM Telco sample dataset (7,043 records), which may not fully represent all telecom customer segments, regional billing patterns, or enterprise-scale data volumes.
- The retention offer recommendation engine uses rule-based logic. A collaborative filtering or reinforcement learning approach would generate more granular, personalised recommendations.
- User credentials are stored in a flat JSON file. A dedicated users table in PostgreSQL would be more robust and auditable at enterprise scale.
- The Streamlit dashboard does not support real-time data ingestion. The ETL pipeline must be re-run manually to refresh customer scores and recommendations.

- PySpark runs in local mode in the current implementation. A distributed cluster deployment (AWS EMR, Databricks, or Azure HDInsight) is required for true enterprise scalability.
- The system currently has no email or SMS notification capability to alert the retention team when a customer crosses the high-risk threshold.

5.3 Future Enhancements

- Apache Airflow integration: automate ETL pipeline scheduling using DAGs for daily or weekly data refresh without manual intervention.
- Real-time streaming ETL: integrate Apache Kafka for real-time customer event ingestion and immediate churn risk recalculation.
- Advanced ML models: implement neural networks (TensorFlow or PyTorch) and AutoML for improved prediction accuracy.
- Cloud deployment: host the pipeline on AWS (S3 + RDS + EC2) or Azure (Blob Storage + Azure SQL + App Service) for production availability.
- SHAP explainability: integrate SHAP values to provide human-readable explanations for each individual churn prediction.
- REST API layer: expose prediction and recommendation endpoints via FastAPI for integration with CRM and billing systems.
- Power BI integration: export scored data to Power BI for enterprise-grade business intelligence and executive reporting.
- Snowflake data warehouse: migrate the analytical storage layer to Snowflake for columnar querying and seamless BI tool integration.

Chapter 6: References / Bibliography

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Chapter 7: Appendices

7.1 Annexure I — Additional Input and Output Screens

The following additional screens are produced by the system and submitted as part of project documentation:

- Screen A1: Streamlit login page with username and password fields, Sign In button, and footer.
- Screen A2: Executive Dashboard — KPI row showing Total Customers (7,043), Churned (1,869), Churn Rate (26.5%), Revenue Lost per Month, Average Tenure, and High-Risk count.
- Screen A3: Churn Predictor form filled with a high-risk customer profile — Month-to-month contract, Electronic Check, Fiber optic, tenure 2 months, monthly charges Rs. 89.
- Screen A4: Churn Predictor output — gauge at 87%, HIGH RISK badge, Retention Call offer card displayed.
- Screen A5: At-Risk Customers page filtered to High risk, showing 10 records with ChurnProbability, Contract, OfferTitle, and MonthlyCharges.
- Screen A6: Model Performance page showing model comparison table and feature importance chart with tenure and MonthlyCharges as top drivers.
- Screen A7: Admin Panel — User Accounts table showing Div-dev (Admin) and at least two regular user accounts with created dates.
- Screen A8: Admin Panel — Access Log table showing at least 10 recent events including login, page views, and prediction runs.

All screens were captured from the running Streamlit application with valid data loaded from the ETL pipeline output (churn_scored.csv, 7,043 records).

7.2 Annexure II — Progress Sheet

Phase	Task Description	Status	Week
1	Project proposal, literature review, and problem definition	Completed	1
2	Environment setup: Python venv, PostgreSQL, VS Code, Git	Completed	2
3	Dataset acquisition from Kaggle and initial EDA (Notebook 01)	Completed	2
4	ETL Phase 1: extract.py — CSV reading, validation, backup	Completed	3
5	ETL Phase 2: transform.py — cleaning, encoding, feature engineering	Completed	4
6	ETL Phase 3: load.py — PostgreSQL schema and data loading	Completed	4
7	PySpark integration: spark_etl.py and distributed pipeline	Completed	5
8	ML pipeline: train_model.py — RF, GBT, LR training and selection	Completed	6
9	Prediction and offer modules: predict.py, recommend.py	Completed	7
10	Streamlit dashboard: app.py — 5 pages with Plotly charts	Completed	8-9
11	Authentication system: auth.py, users.json, access_log.csv	Completed	10
12	Unit testing: test_etl.py — 9 classes, 35 test cases, 100% pass	Completed	11
13	Jupyter notebooks 01-04: EDA, cleaning, features, visualisation	Completed	11
14	UML diagrams (9 figures) and project report writing	Completed	12
15	Final review, screenshot capture, and submission preparation	Completed	13